

Day 4 Homework task 2: ChromaDB filter exercise

Pvm: 12.04.2026

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Tehtävä 1

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# -----  
# EXERCISE 1 – Metadata filtering with where  
# -----  
# TODO: Use collection.get() with a where filter to fetch all documents  
# where category == "vpn"  
#  
# Expected: 3 documents (doc-000, doc-001, doc-002)  
  
print("\n--- EXERCISE 1: Basic metadata filter ---")  
  
# Write your code here:  
#collection.get(where={"category": "vpn"})  
print(collection.get(where={"category": "vpn"}))  
  
"
```

Output

```
--- EXERCISE 1: Basic metadata filter ---  
{'ids': ['doc-000', 'doc-001', 'doc-002'], 'embeddings': None, 'documents': ['To connect to the university VPN, install the GlobalProtect client from software.uni.fi and authenticate with your student credentials.', 'The VPN service is maintained every Tuesday between 06:00 and 07:00. Connections will be interrupted during this window.', 'If the VPN disconnects repeatedly, try switching the server region from EU-West to EU-North in GlobalProtect settings.'], 'uris': None, 'included': ['metadatas', 'documents'], 'data': None, 'metadatas': [{'verified': True, 'year': 2025, 'category': 'vpn', 'priority': 'high'}, {'category': 'vpn', 'year': 2025, 'verified': True, 'priority': 'medium'}, {'category': 'vpn', 'priority': 'low', 'year': 2024, 'verified': True}]}
```

Tehtävä 2

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# -----  
# EXERCISE 2 – Combining filters with $and and $or  
# -----  
# TODO: Use collection.get() with a combined where filter to find documents  
# where priority == "high" AND year == 2025 AND verified == True  
#  
# Expected: 7 documents covering vpn, email, network, accounts categories  
  
print("\n--- EXERCISE 2: Combined metadata filters ---")  
  
# Write your code here:  
print(collection.get(where={"$and":  
    [{"priority": "high"},  
    {"year": 2025},  
    {"verified": True}]}))  
  
print("\n",collection.get(where={"$or":  
    [{"category": "software"},  
    {"category": "printing"}]}))
```

Output

```
--- EXERCISE 2: Combined metadata filters ---  
{'ids': ['doc-000', 'doc-003', 'doc-009', 'doc-010', 'doc-012', 'doc-013', 'doc-014'], 'embeddings': None, 'documents': ['To connect to the university VPN, install the GlobalProtect client from software.uni.fi and authenticate with your student credentials.', 'Student email accounts are hosted on Microsoft 365. Your address is firstname.lastname@student.uni.fi.', 'The campus Wi-Fi network is called UniMet. Use your student credentials to authenticate via the captive portal.', 'Eduroam is available on campus and at partner universities worldwide. Configure it with your full student email address.', 'Passwords must be at least 12 characters and include uppercase, lowercase, a digit, and a special character.', 'Your student account is created within 24 hours of enrolment confirmation. Check your personal email for the activation link.', 'Multi-factor authentication (MFA) is mandatory for all student accounts from September 2025 onwards.'], 'uris': None, 'included': ['metadata', 'documents'], 'data': None, 'metadata': [{'year': 2025, 'priority': 'high', 'verified': True, 'category': 'vpn'}, {'year': 2025, 'priority': 'high', 'category': 'email', 'verified': True}, {'year': 2025, 'category': 'network', 'priority': 'high', 'verified': True}, {'year': 2025, 'verified': True, 'priority': 'high', 'category': 'network'}, {'priority': 'high', 'category': 'accounts', 'year': 2025, 'verified': True}, {'category': 'accounts', 'year': 2025, 'verified': True, 'priority': 'high'}, {'priority': 'high', 'category': 'accounts', 'verified': True, 'year': 2025}]}  
  
{'ids': ['doc-006', 'doc-007', 'doc-008', 'doc-015', 'doc-016', 'doc-017'], 'embeddings': None, 'documents': ['MATLAB is available for all students via the campus licence. Download it from software.uni.fi using your student ID.', 'Microsoft Office 365 is included in your student account at no cost. Install up to five devices.', 'Adobe Creative Cloud is available for ICT and Design students only. Request access through the IT helpdesk portal.', 'Campus printer accepts UniPrint credits. Load credits at any campus info desk or online at print.uni.fi.', 'The default print quota is 200 pages per semester. ICT students receive an additional 100 pages.', 'Colour printing costs 0.15 EUR per page. Black-and-white printing costs 0.04 EUR per page.'], 'uris': None, 'included': ['metadata', 'documents'], 'data': None, 'metadata': [{'priority': 'medium', 'category': 'software', 'verified': True, 'year': 2025}, {'priority': 'low', 'year': 2025, 'category': 'software', 'verified': True}, {'priority': 'low', 'verified': False, 'year': 2024, 'category': 'software'}, {'priority': 'medium', 'year': 2025, 'verified': True, 'category': 'printing'}, {'priority': 'low', 'year': 2025, 'verified': True, 'category': 'printing'}, {'priority': 'low', 'category': 'printing', 'year': 2025, 'verified': True}]}
```

Tehtävä 3

```
# -----  
# EXERCISE 3 – Full text search with where_document  
# -----  
# TODO A: Use collection.get() with where_document to find all documents  
#         that contain the word "student"  
#  
# TODO B: Extend it – find documents that contain "student" but do NOT  
#         contain "password" (use $and with $contains and $not_contains)  
  
print("\n--- EXERCISE 3: Full text search ---")  
  
# Write your code here:  
print(collection.get(where_document={"$contains": "student"}))  
print("\n",collection.get(where_document={"$and":  
    [{"$contains": "student"},  
    {"$not_contains": "password"}]}))  
  
#
```

Output

```
--- EXERCISE 3: Full text search ---  
{'ids': ['doc-000', 'doc-001', 'doc-006', 'doc-007', 'doc-008', 'doc-009', 'doc-010', 'doc-013', 'doc-014', 'doc-016'], 'embeddings': None, 'documents': ['To connect to the university VPN, install the GlobalProtect client from software.uni.fi and authenticate with your student credentials.', 'Student email accounts are hosted on Microsoft 365. Your address is firstname.lastname@student.uni.fi.', 'MATLAB is available for all students via the campus licence. Download it from software.uni.fi using your student ID.', 'Microsoft Office 365 is included in your student account at no cost. Install up to five devices.', 'Adobe Creative Cloud is available for ICT and Design students only. Request access through the IT helpdesk portal.', 'The campus Wi-Fi network is called UniNet. Use your student credentials to authenticate via the captive portal.', 'Eduroam is available on campus and at partner universities worldwide. Configure it with your full student email address.', 'Your student account is created within 24 hours of enrolment confirmation. Check your personal email for the activation link.', 'Multi-factor authentication (MFA) is mandatory for all student accounts from September 2025 onwards.', 'The default print quota is 200 pages per semester. ICT students receive an additional 100 pages.'], 'uris': None, 'included': ['metadata', 'documents'], 'data': None, 'metadata': [{'priority': 'high', 'year': 2025, 'verified': true, 'category': 'vpn'}, {'verified': true, 'category': 'email', 'priority': 'high', 'year': 2025}, {'category': 'software', 'priority': 'medium', 'verified': true, 'year': 2025}, {'verified': true, 'category': 'software', 'priority': 'low', 'year': 2025}, {'verified': false, 'category': 'software', 'priority': 'low', 'year': 2024}, {'year': 2025, 'priority': 'high', 'category': 'network', 'verified': true}, {'verified': true, 'year': 2025, 'category': 'network', 'priority': 'high'}, {'year': 2025, 'category': 'accounts', 'verified': true, 'priority': 'high'}, {'verified': true, 'priority': 'high', 'year': 2025, 'category': 'accounts'}, {'priority': 'low', 'year': 2025, 'category': 'printing', 'verified': true}]]  
  
{'ids': ['doc-000', 'doc-001', 'doc-006', 'doc-007', 'doc-008', 'doc-009', 'doc-010', 'doc-013', 'doc-014', 'doc-016'], 'embeddings': None, 'documents': ['To connect to the university VPN, install the GlobalProtect client from software.uni.fi and authenticate with your student credentials.', 'Student email accounts are hosted on Microsoft 365. Your address is firstname.lastname@student.uni.fi.', 'MATLAB is available for all students via the campus licence. Download it from software.uni.fi using your student ID.', 'Microsoft Office 365 is included in your student account at no cost. Install up to five devices.', 'Adobe Creative Cloud is available for ICT and Design students only. Request access through the IT helpdesk portal.', 'The campus Wi-Fi network is called UniNet. Use your student credentials to authenticate via the captive portal.', 'Eduroam is available on campus and at partner universities worldwide. Configure it with your full student email address.', 'Your student account is created within 24 hours of enrolment confirmation. Check your personal email for the activation link.', 'Multi-factor authentication (MFA) is mandatory for all student accounts from September 2025 onwards.', 'The default print quota is 200 pages per semester. ICT students receive an additional 100 pages.'], 'uris': None, 'included': ['metadata', 'documents'], 'data': None, 'metadata': [{'year': 2025, 'priority': 'high', 'category': 'vpn', 'verified': true}, {'priority': 'high', 'category': 'email', 'verified': true, 'year': 2025}, {'year': 2025, 'category': 'software', 'priority': 'medium', 'verified': true}, {'year': 2025, 'category': 'software', 'verified': true, 'priority': 'low'}, {'year': 2024, 'priority': 'low', 'category': 'software', 'verified': false}, {'priority': 'high', 'category': 'network', 'verified': true, 'year': 2025}, {'category': 'network', 'verified': true, 'priority': 'high', 'year': 2025}, {'verified': true, 'priority': 'high', 'category': 'accounts', 'year': 2025}, {'year': 2025, 'verified': true, 'category': 'accounts', 'priority': 'high'}, {'verified': true, 'priority': 'low', 'category': 'printing', 'year': 2025}]]
```

Tehtävä 4

```
# -----
# EXERCISE 4 – Combining semantic query with metadata and text filters
# -----
# TODO: Use collection.query() to find documents semantically related to
#       "how do I print documents on campus"
#       BUT restrict results to category == "printing" using where,
#       AND only include documents that contain the word "page" using where_document
#
# Observe: how do the distances change compared to an unfiltered query?

print("\n--- EXERCISE 4: Semantic query + metadata filter + text filter ---")

# Write your code here:

print(collection.query(
    query_texts=["how do I print documents on campus"],
    n_results=6,
    where={"category": "printing"},
    where_document={"$contains": "page"}
))
```

Output

```
--- EXERCISE 4: Semantic query + metadata filter + text filter ---
{'ids': [['doc-016', 'doc-017']], 'embeddings': None, 'documents': [['The default print quota is 200 pages per semester. ICT students receive an additional 100 pages.', 'Colour printing costs 0.15 EUR per page. Black and white printing costs 0.04 EUR per page.'], ['The default print quota is 200 pages per semester. ICT students receive an additional 100 pages.', 'Colour printing costs 0.15 EUR per page. Black and white printing costs 0.04 EUR per page.']], 'uris': None, 'included': ['metadatas', 'documents', 'distances'], 'data': None, 'metadatas': [[{'year': 2025, 'verified': True, 'priority': 'low', 'category': 'printing'}, {'year': 2025, 'category': 'printing', 'verified': True}], 'distances': [[1.0045626163482666, 1.4034862518310542]]]
(verbose) PS D:\Github\llm-ai-rag-examples\4.2-homework-started-chromadb-filter>
```